



Dental and Craniofacial Anomalies



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There are many developmental and acquired anomalies that can affect the teeth and facial skeleton so they classified into:

- ☐ **Anomalies of the teeth**
- ☐ **Craniofacial anomalies.**

The diagnosis of them based on both the **clinical** and **radiographic** findings.





1- Developmental teeth anomalies: Number, Structure, Size, Shape (morphology), and Eruption.

2-Acquired teeth Abnormalities:Attrition,Abrasion, Erosion, Pulp stones





Missing teeth (Hypodontia) It's a condition at which the patient has missing one or few teeth as a result of their failure to develop.

(oligodontia) The term for when numerous teeth are absent
(anodontia). The term for the failure of all teeth to develop

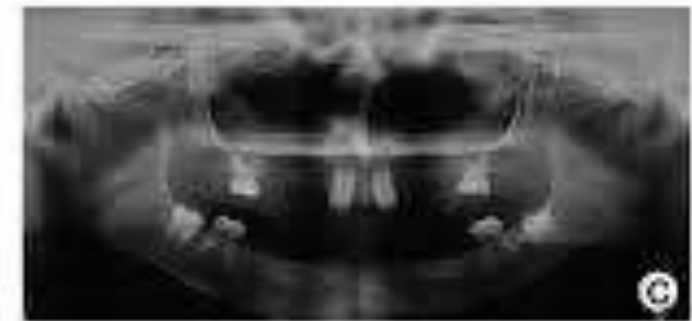


Imaging features of Missing teeth may be recognized by **identifying and counting the teeth present.**





It could happen in
Ectodermal Dysplasia.





Hyperdontia It's a condition of having supernumerary teeth.

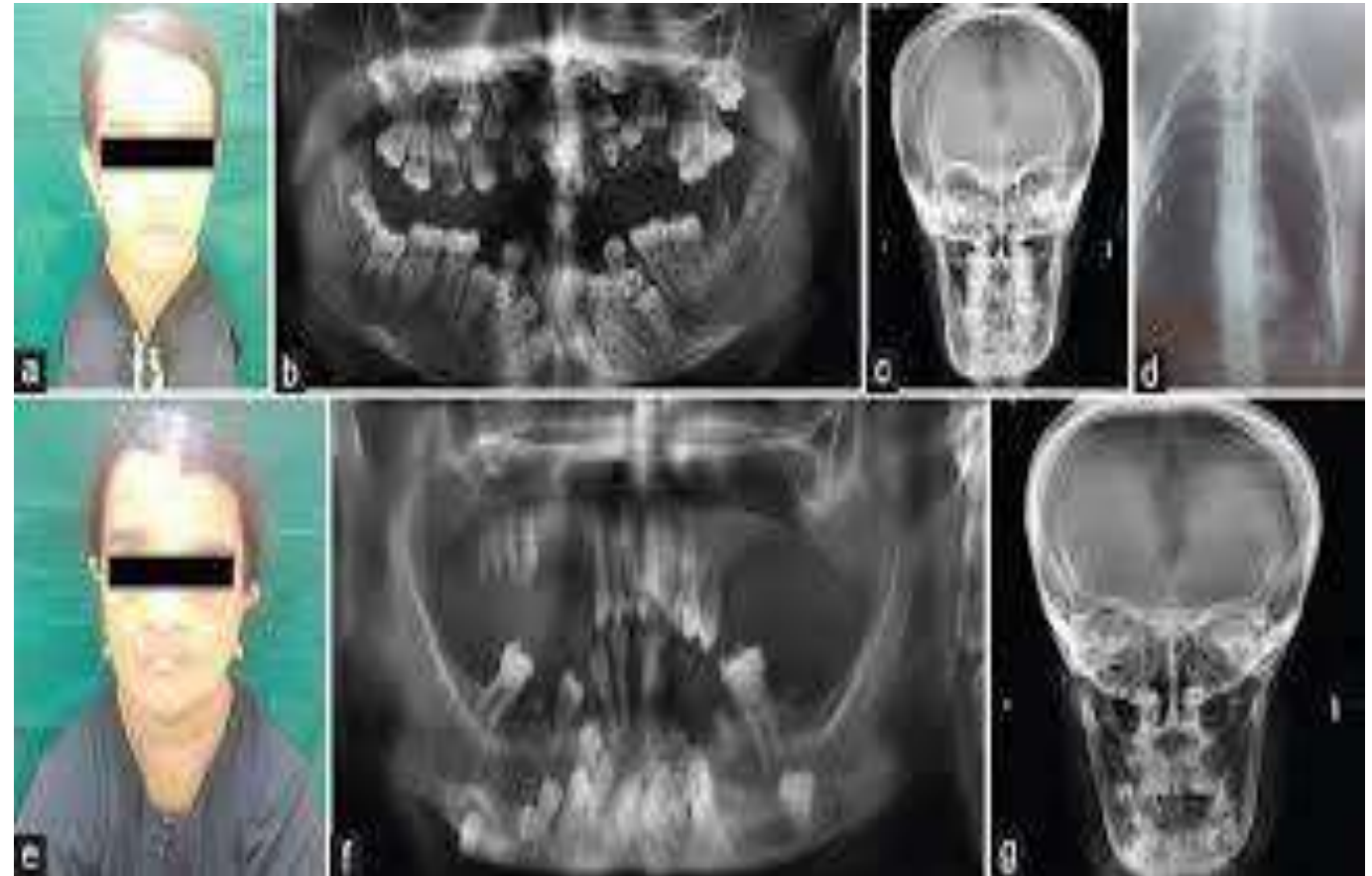
The most common supernumerary tooth is **a mesiodens**, which is a malformed, peg-like tooth that occurs between the maxillary central incisors, Fourth and fifth molars (**paramolar, or distomolar**) that form behind the third molars are another kind of supernumerary teeth.

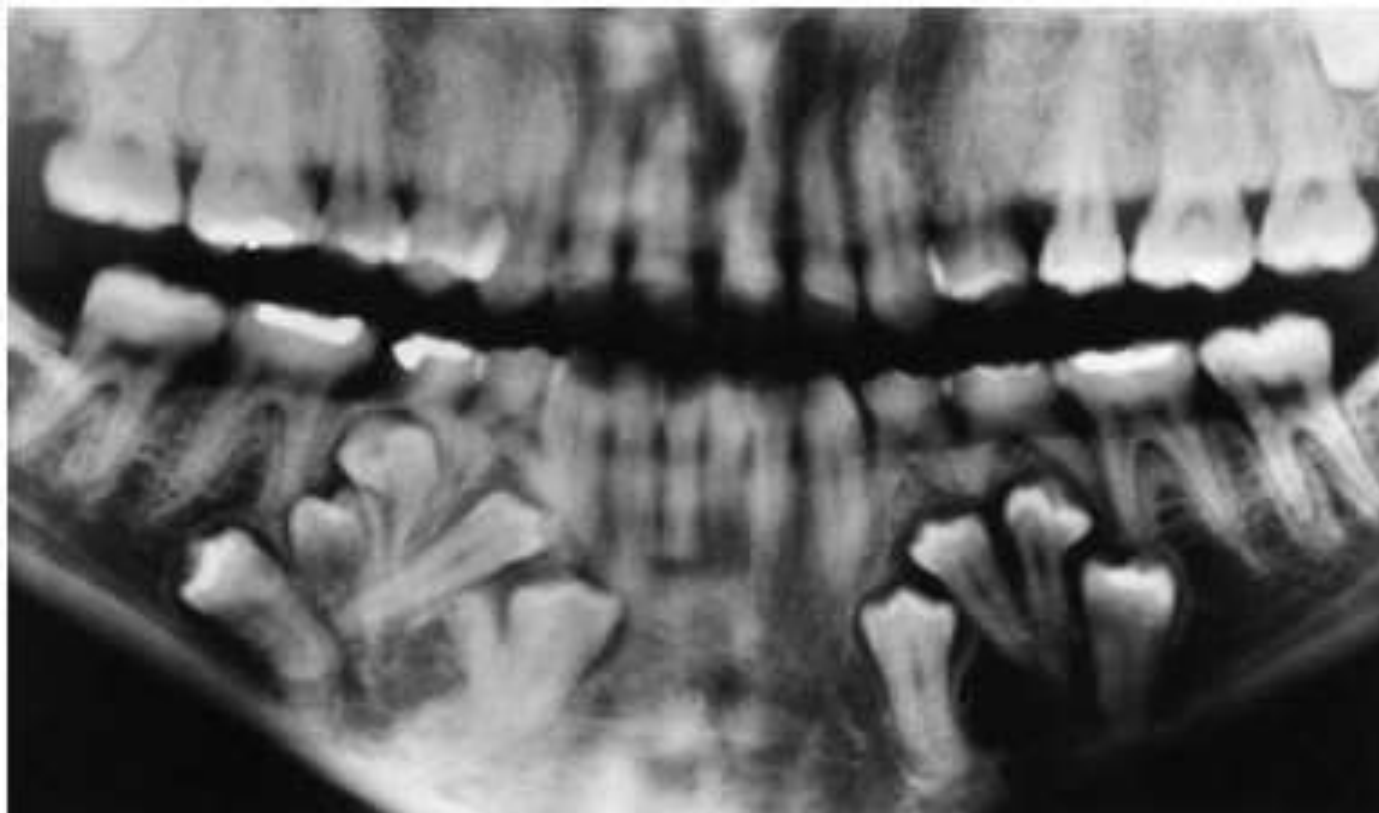
The imaging features of supernumerary teeth are variable. They may appear entirely **normal in both size and shape** or they may be **smaller and conical shape**.





It could be associated with
Cleidocranial Dysplasia.





3.18 Mesiodens. Supernumerary tooth |





The Abnormalities in teeth structures are subdivided into:

1- Genetic defect:

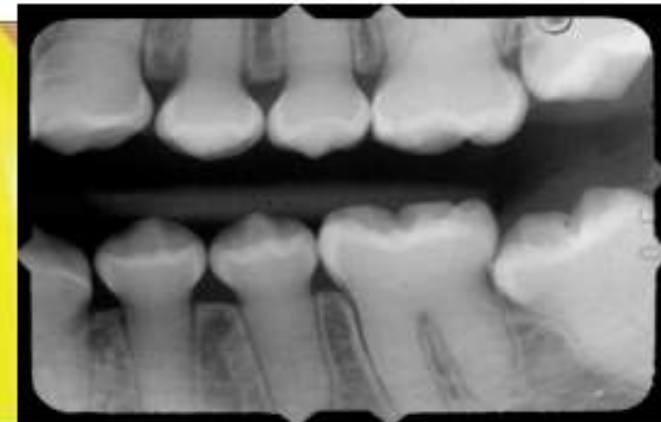
2- Acquired defects:

Amelogenesis imperfecta Genetic disturbances in enamel formation leading to altered morphology of enamel. There is normal dentin and pulp formation. Imaging features shows square-shaped crown, thin enamel and absence of cusps,





Dentinogenesis imperfecta Is a genetic anomaly involving the dentin in both deciduous and permanent dentition **Imaging features** show a marked constriction at cervical region of the tooth crown with **pulp chamber obliteration** and **short blunt roots. (bulbous)**





Regional Odontodysplasia (ghost teeth) Its a **rare** condition in which both enamel and dentin are hypoplastic and hypocalcified. Imaging feature described as “ghost-like” appearance. **The pulp chambers are large and the root canals are wide with very thin enamel and dentin.**



Fig. 19.2. Regional odontodysplasia: ghost teeth.





Turner hypoplasia It's a frequent pattern of enamel defects seen in permanent teeth secondary to periapical inflammatory disease of the overlying deciduous tooth. The altered tooth is called (Turner's tooth).

Imaging features of the involved region of the crown may appear as an ill-defined radiolucent region.





Congenital syphilis It's a dental hypoplasia that results from direct infection of the developing tooth by spirochete of syphilis, involves the permanent incisors that called (**Hutchinson's teeth**) and first molars that called (**mulberry molars**). **Imaging features** have a characteristic shapes of the affected incisor and molar crowns.



Hutchinson's teeth





Macrodontia (large teeth) It's a condition in which the teeth are abnormally large, rarely affects the entire dentition. Often a single tooth or a group of teeth may be involved. **Imaging features** reveal the increased size of the teeth. The shape of the tooth is usually normal, but some cases may be distorted. It associated with crowding, malocclusion, or impaction.





Microdontia (small teeth) It's a condition in which teeth appear smaller than normal. In the generalized form, all teeth are involved. In the localized form, only single or few teeth are involved. The most common teeth affected are the upper lateral incisors and third molars. **Imaging features** of the affected teeth are frequently small and malformed.





Fusion Two teeth joined together into a single anatomic crown (union of two separated tooth germ). Fusion is more common in anterior teeth of both the deciduous and permanent dentitions. **Imaging features** of fused teeth show unusual shape, size, and pulp chamber or root canal



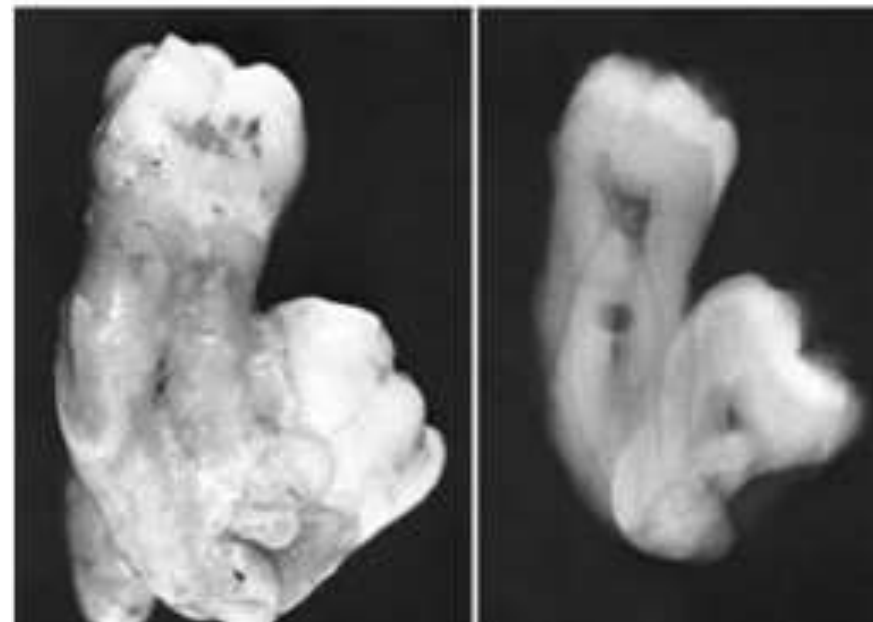


Gemination Single tooth germ divided into two teeth joined together (single root with two or enlarged crowns). **Imaging features** reveal the altered shape of the hard tissue and pulp chamber of the geminated tooth.





Concrescence Is union of two adjacent teeth by cementum only. Maxillary third molar frequently involved. **Imaging features** reveal concrescence teeth may be in close contact or are simply superimposed.





- **Dilaceration** Is a deviation or sharp bend in the linear relationship of a tooth crown to its root; **Imaging features** is readily apparent on an intraoral radiograph when the roots are bending mesially or distally, buccally or lingually





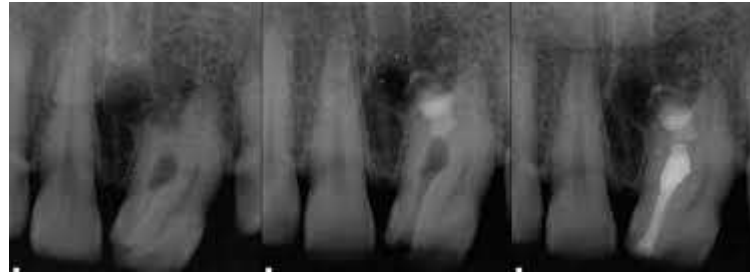
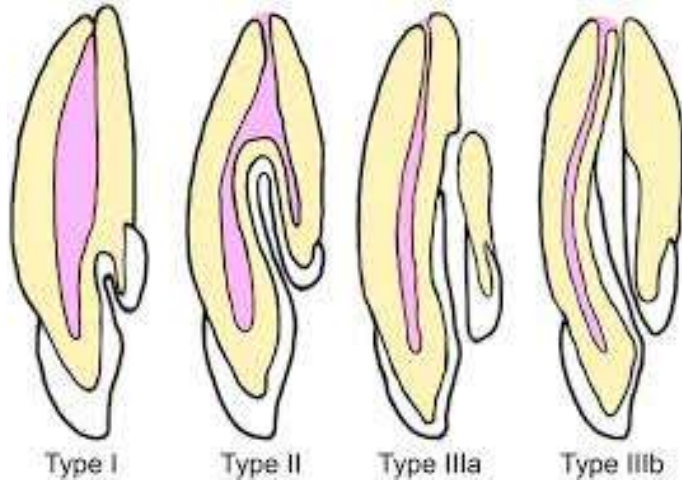
Dens Invaginatus and Dens Evaginatus Dens invaginatus is infolding of the enamel surface into the interior of a tooth crown or the root .

Imaging features show more radiopaque than the surrounding tooth structure, **poorly defined root**,

while Dens evaginatus is the outpouching of the enamel on the occlusal surface.

Imaging features shows an extension of a dentin tubercle on the occlusal surface covered with radiopaque enamel.





Dens Invaginatus



Dens Evaginatus





Enamel pearl It's small spherical enamel masses (enameloma) located at the root of the molars and are found in 3% of the population.

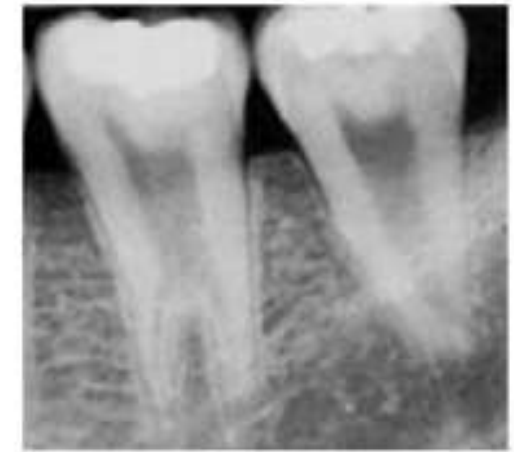
Imaging features of the enamel pearl appears smooth, round radiopaque structure.





Taurodontism It's a condition found in the molar teeth whereby the body of the tooth and pulp chamber is enlarged vertically with short roots, the floor of the pulp and the furcation of the tooth is moved apically.

Imaging features is the elongated pulp chamber and the more apically positioned furcation, shortened roots with long crown.





Talon cusp: It's an Accessory cusp like structure projecting from the cingulum area or cementenamel junction of the maxillary or mandibular anterior teeth.

Imaging features show a distinct radiopaque image of talon cusp on the crown of the involved tooth.





Eruption of Teeth:

Transposition is the condition in which two adjacent teeth have exchanged positions in the dental arch.

Imaging features reveal transposition when the teeth are not in their usual sequence in the dental arch





Premature Eruption (natal and neonatal teeth)

The teeth erupted in the oral cavity at the time of birth are called as '**natal teeth**' and teeth erupting prematurely in first 30 days of life are called as '**neonatal teeth**'.

Imaging features the roots are not seen on the radiograph and the teeth are very small.





Delayed Eruption (Impacted Teeth)

permanent teeth are observed to be delayed in eruption forming Partially or completely impacted teeth, is more commonly in **mandibular third molar**, followed by **maxillary canine** and **maxillary third molar**.

Imaging features of impacted mandibular third molar show mesioangular , distoangular , vertical or horizontal impaction.





2. Acquired Teeth Abnormalities

Attrition

The physiologic (normal) wearing of teeth, typically from occlusal (chewing) contacts

Imaging Appearance: A flat incisal or occlusal surface.

Abrasion

The non-physiologic (abnormal) wearing of teeth caused by friction with a foreign object, such as a toothbrush, dental floss, or pipe

Imaging Appearance: Defects are seen at the cervical (neck) level of the teeth

Erosion

The loss of tooth structure due to chemical action.

Imaging Features: Appears as radiolucent (dark) defects on the crown.

Resorption

The removal of tooth structure, which can be internal or external

Imaging (External Resorption): Shows a smooth loss of structure at the apical (tip) and cervical regions , often with a blunt root apex

Imaging (Internal Resorption) : Appears as a round or oval radiolucent (dark) area within the root or crown

Secondary Dentin

Additional dentin that is deposited in the tooth.

Imaging Features: A reduction in the size of the normal pulp chamber.

Pulp Stone

Foci (small areas) of calcification within the dental pulp.

Imaging Appearance: Appear as radiopaque (white) structures inside the pulp.

Pulpal Sclerosis

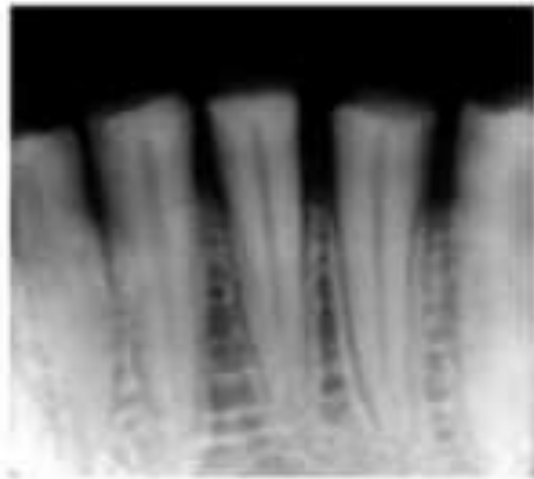
Another form of pulp calcification, but it is diffused (spread out) over a larger area.

Hypercementosis

The excessive deposition of cementum on the roots of a tooth.

Imaging Appearance: An excessive buildup of cementum around all or part of a root.





Attrition



abrasion



internal resorption



pulp stone





Imaging Features appeared as well-defined vertical radiolucent defect in the alveolar process with numerous dental anomalies (absence of the maxillary lateral incisor, presence of supernumerary teeth, enamel hypoplasia, malformed teeth, delay eruption with hypodontia in both arches).

